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ECE 3317

Homework #2

Assigned: Monday Feb 9th

Due: Wed Feb 18th

1) If $\underline{B}(x, y, z) = xy^2\hat{x} + xy^2z^3\hat{y} + x^2yz^2\hat{z}$, find $\underbrace{\nabla \times \underline{B}}_{\text{Curl}}$ and $\underbrace{\nabla \cdot \underline{B}}_{\text{Divergence}}$.

$$\nabla = \hat{x} \frac{\partial}{\partial x} + \hat{y} \frac{\partial}{\partial y} + \hat{z} \frac{\partial}{\partial z}$$

$$\nabla \times \underline{B} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xy^2 & xy^2z^3 & x^2yz^2 \end{vmatrix} = \hat{x} \left[\frac{\partial}{\partial y}(x^2yz^2) - \frac{\partial}{\partial z}(xy^2z^3) \right] + \hat{y} \left[\frac{\partial}{\partial z}(xy^2) - \frac{\partial}{\partial x}(x^2yz^2) \right] + \hat{z} \left[\frac{\partial}{\partial x}(xy^2z^3) - \frac{\partial}{\partial y}(xy^2) \right]$$

$$\nabla \times \underline{B} = \hat{x} [x^2z^2 - 3xy^2z^2] + \hat{y} [0 - 2xyz^2] + \hat{z} [y^2z^3 - 2xy]$$

$$\nabla \times \underline{B} = (x^2z^2 - 3xy^2z^2)\hat{x} - 2xyz^2\hat{y} + (y^2z^3 - 2xy)\hat{z}$$

$$\nabla \cdot \underline{B} = \frac{\partial}{\partial x}(xy^2) + \frac{\partial}{\partial y}(xy^2z^3) + \frac{\partial}{\partial z}(x^2yz^2) = y^2 + 2xyz^3 + 2x^2yz$$

$$\nabla \cdot \underline{B} = y^2 + 2xyz^3 + 2x^2yz$$

2) Prove that for an arbitrary vector $\underline{A}(x, y, z)$ and an arbitrary scalar function $\phi(x, y, z)$, that

$$\nabla \cdot (\nabla \times \underline{A}) = 0 \quad \underline{A}(x, y, z) = A_x \underline{\hat{x}} + A_y \underline{\hat{y}} + A_z \underline{\hat{z}}$$

$$\nabla \times (\nabla \phi) = \underline{0}.$$

Hint: It is easiest if you use rectangular coordinates to prove these identities. (But keep in mind that if you prove them in any coordinate system, the identities must be true in all coordinate systems.)

$$\text{Curl: } \nabla \times \underline{A} = \underline{\hat{x}} \left[\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right] + \underline{\hat{y}} \left[\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right] + \underline{\hat{z}} \left[\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right]$$

$$\nabla = \underline{\hat{x}} \frac{\partial}{\partial x} + \underline{\hat{y}} \frac{\partial}{\partial y} + \underline{\hat{z}} \frac{\partial}{\partial z}$$

$$\nabla \cdot (\nabla \times \underline{A}) = \frac{\partial}{\partial x} \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) + \frac{\partial}{\partial y} \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) + \frac{\partial}{\partial z} \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right)$$

$$\nabla \cdot (\nabla \times \underline{A}) = \frac{\partial^2 A_z}{\partial x \partial y} - \frac{\partial^2 A_y}{\partial x \partial z} + \frac{\partial^2 A_x}{\partial y \partial z} - \frac{\partial^2 A_z}{\partial x \partial y} + \frac{\partial^2 A_y}{\partial x \partial z} - \frac{\partial^2 A_x}{\partial y \partial z}$$

$$\nabla \cdot (\nabla \times \underline{A}) = \left(\frac{\partial^2 A_z}{\partial x \partial y} - \frac{\partial^2 A_z}{\partial x \partial y} \right) + \left(\frac{\partial^2 A_x}{\partial y \partial z} - \frac{\partial^2 A_x}{\partial y \partial z} \right) + \left(\frac{\partial^2 A_y}{\partial x \partial z} - \frac{\partial^2 A_y}{\partial x \partial z} \right)$$

$$\nabla \cdot (\nabla \times \underline{A}) = 0 + 0 + 0 = 0$$

$$\text{Gradient: } \nabla \phi = \underline{\hat{x}} \left(\frac{\partial \phi}{\partial x} \right) + \underline{\hat{y}} \left(\frac{\partial \phi}{\partial y} \right) + \underline{\hat{z}} \left(\frac{\partial \phi}{\partial z} \right)$$

$$\nabla \times (\nabla \phi) = \begin{vmatrix} \underline{\hat{x}} & \underline{\hat{y}} & \underline{\hat{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial \phi}{\partial x} & \frac{\partial \phi}{\partial y} & \frac{\partial \phi}{\partial z} \end{vmatrix} = \underline{\hat{x}} \left[\frac{\partial}{\partial y} \left(\frac{\partial \phi}{\partial z} \right) - \frac{\partial}{\partial z} \left(\frac{\partial \phi}{\partial y} \right) \right] + \underline{\hat{y}} \left[\frac{\partial}{\partial z} \left(\frac{\partial \phi}{\partial x} \right) - \frac{\partial}{\partial x} \left(\frac{\partial \phi}{\partial z} \right) \right] + \underline{\hat{z}} \left[\frac{\partial}{\partial x} \left(\frac{\partial \phi}{\partial y} \right) - \frac{\partial}{\partial y} \left(\frac{\partial \phi}{\partial x} \right) \right]$$

$$\nabla \times (\nabla \phi) = \underline{\hat{x}} \left[\frac{\partial^2 \phi}{\partial y \partial z} - \frac{\partial^2 \phi}{\partial y \partial z} \right] + \underline{\hat{y}} \left[\frac{\partial^2 \phi}{\partial x \partial z} - \frac{\partial^2 \phi}{\partial x \partial z} \right] + \underline{\hat{z}} \left[\frac{\partial^2 \phi}{\partial x \partial y} - \frac{\partial^2 \phi}{\partial x \partial y} \right] = \underline{\hat{x}}(0) + \underline{\hat{y}}(0) + \underline{\hat{z}}(0)$$

$$\nabla \times (\nabla \phi) = \underline{0}$$

3) Recall that the vector Laplacian of a vector function $\underline{V}(x, y, z)$ is defined as

$$\nabla^2 \underline{V} \equiv \nabla(\nabla \cdot \underline{V}) - \nabla \times (\nabla \times \underline{V}).$$

Prove that in rectangular coordinates the following identity holds:

$$\nabla^2 \underline{V} = \hat{x} \nabla^2 V_x + \hat{y} \nabla^2 V_y + \hat{z} \nabla^2 V_z.$$

In words, this identity says that the rectangular components of the vector Laplacian are equal to the scalar Laplacian of the corresponding components of the vector function. This identity only holds in rectangular coordinates, not in cylindrical or spherical coordinates.

(The vector Laplacian and the identity that you are proving above are both very useful when deriving the scalar Helmholtz equation for the components of the electric and magnetic fields, something we will be doing later.)

$$\nabla = \hat{x} \frac{\partial}{\partial x} + \hat{y} \frac{\partial}{\partial y} + \hat{z} \frac{\partial}{\partial z}, \quad \underline{V} = V_x \hat{x} + V_y \hat{y} + V_z \hat{z}$$

$$\nabla^2 \underline{V} \equiv \nabla(\nabla \cdot \underline{V}) - \nabla \times (\nabla \times \underline{V})$$

$$\nabla \cdot \underline{V} = \frac{\partial}{\partial x}(V_x) + \frac{\partial}{\partial y}(V_y) + \frac{\partial}{\partial z}(V_z) = \frac{\partial V_x}{\partial x} + \frac{\partial V_y}{\partial y} + \frac{\partial V_z}{\partial z}$$

$$\nabla \cdot \underline{V} = \frac{\partial V_x}{\partial x} + \frac{\partial V_y}{\partial y} + \frac{\partial V_z}{\partial z}$$

$$\nabla(\nabla \cdot \underline{V}) = \hat{x} \left(\frac{\partial}{\partial x} \left(\frac{\partial V_x}{\partial x} + \frac{\partial V_y}{\partial y} + \frac{\partial V_z}{\partial z} \right) \right) + \hat{y} \left(\frac{\partial}{\partial y} \left(\frac{\partial V_x}{\partial x} + \frac{\partial V_y}{\partial y} + \frac{\partial V_z}{\partial z} \right) \right) + \hat{z} \left(\frac{\partial}{\partial z} \left(\frac{\partial V_x}{\partial x} + \frac{\partial V_y}{\partial y} + \frac{\partial V_z}{\partial z} \right) \right)$$

$$\nabla(\nabla \cdot \underline{V}) = \hat{x} \left[\frac{\partial^2 V_x}{\partial x^2} + \frac{\partial^2 V_y}{\partial x \partial y} + \frac{\partial^2 V_z}{\partial x \partial z} \right] + \hat{y} \left[\frac{\partial^2 V_x}{\partial x \partial y} + \frac{\partial^2 V_y}{\partial y^2} + \frac{\partial^2 V_z}{\partial y \partial z} \right] + \hat{z} \left[\frac{\partial^2 V_x}{\partial x \partial z} + \frac{\partial^2 V_y}{\partial y \partial z} + \frac{\partial^2 V_z}{\partial z^2} \right]$$

$$\nabla \times \underline{V} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ V_x & V_y & V_z \end{vmatrix} = \hat{x} \left[\frac{\partial V_z}{\partial y} - \frac{\partial V_y}{\partial z} \right] + \hat{y} \left[\frac{\partial V_x}{\partial z} - \frac{\partial V_z}{\partial x} \right] + \hat{z} \left[\frac{\partial V_y}{\partial x} - \frac{\partial V_x}{\partial y} \right]$$

$$\nabla \times (\nabla \times \underline{V}) = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial V_z}{\partial y} - \frac{\partial V_y}{\partial z} & \frac{\partial V_x}{\partial z} - \frac{\partial V_z}{\partial x} & \frac{\partial V_y}{\partial x} - \frac{\partial V_x}{\partial y} \end{vmatrix} = \hat{x} \left[\frac{\partial}{\partial y} \left(\frac{\partial V_y}{\partial z} - \frac{\partial V_z}{\partial y} \right) - \frac{\partial}{\partial z} \left(\frac{\partial V_x}{\partial z} - \frac{\partial V_z}{\partial x} \right) \right] + \hat{y} \left[\frac{\partial}{\partial z} \left(\frac{\partial V_z}{\partial y} - \frac{\partial V_y}{\partial z} \right) - \frac{\partial}{\partial x} \left(\frac{\partial V_y}{\partial x} - \frac{\partial V_x}{\partial y} \right) \right] + \hat{z} \left[\frac{\partial}{\partial x} \left(\frac{\partial V_x}{\partial z} - \frac{\partial V_z}{\partial x} \right) - \frac{\partial}{\partial y} \left(\frac{\partial V_z}{\partial y} - \frac{\partial V_y}{\partial z} \right) \right]$$

$$\nabla \times (\nabla \times \underline{V}) = \hat{x} \left[\frac{\partial^2 V_x}{\partial x \partial y} - \frac{\partial^2 V_z}{\partial y^2} - \frac{\partial^2 V_x}{\partial z^2} + \frac{\partial^2 V_z}{\partial x \partial z} \right] + \hat{y} \left[\frac{\partial^2 V_z}{\partial y \partial z} - \frac{\partial^2 V_y}{\partial z^2} - \frac{\partial^2 V_y}{\partial x^2} + \frac{\partial^2 V_x}{\partial x \partial y} \right] \\ + \hat{z} \left[\frac{\partial^2 V_x}{\partial x \partial z} - \frac{\partial^2 V_z}{\partial x^2} - \frac{\partial^2 V_z}{\partial y^2} + \frac{\partial^2 V_y}{\partial y \partial z} \right]$$

$$\nabla (\nabla \cdot \underline{V}) - \nabla \times (\nabla \times \underline{V}) = \hat{x} \left[\left(\frac{\partial^2 V_x}{\partial x^2} + \frac{\partial^2 V_y}{\partial x \partial y} + \frac{\partial^2 V_z}{\partial x \partial z} \right) - \left(\frac{\partial^2 V_y}{\partial x \partial y} - \frac{\partial^2 V_z}{\partial y^2} - \frac{\partial^2 V_x}{\partial z^2} + \frac{\partial^2 V_z}{\partial x \partial z} \right) \right] \\ + \hat{y} \left[\left(\frac{\partial^2 V_x}{\partial x \partial y} + \frac{\partial^2 V_y}{\partial y^2} + \frac{\partial^2 V_z}{\partial y \partial z} \right) - \left(\frac{\partial^2 V_z}{\partial y \partial z} - \frac{\partial^2 V_y}{\partial z^2} - \frac{\partial^2 V_y}{\partial x^2} + \frac{\partial^2 V_x}{\partial x \partial y} \right) \right] \\ + \hat{z} \left[\left(\frac{\partial^2 V_x}{\partial x \partial z} + \frac{\partial^2 V_y}{\partial y \partial z} + \frac{\partial^2 V_z}{\partial z^2} \right) - \left(\frac{\partial^2 V_x}{\partial x \partial z} - \frac{\partial^2 V_z}{\partial x^2} - \frac{\partial^2 V_z}{\partial y^2} + \frac{\partial^2 V_y}{\partial y \partial z} \right) \right]$$

$$\nabla \cdot (\nabla \cdot \underline{V}) - \nabla \times (\nabla \times \underline{V}) = \hat{x} \left[\frac{\partial^2 V_x}{\partial x^2} + \frac{\partial^2 V_x}{\partial y^2} + \frac{\partial^2 V_x}{\partial z^2} \right] + \hat{y} \left[\frac{\partial^2 V_y}{\partial x^2} + \frac{\partial^2 V_y}{\partial y^2} + \frac{\partial^2 V_y}{\partial z^2} \right] \\ + \hat{z} \left[\frac{\partial^2 V_z}{\partial x^2} + \frac{\partial^2 V_z}{\partial y^2} + \frac{\partial^2 V_z}{\partial z^2} \right]$$

$$\nabla \cdot (\nabla \cdot \underline{V}) - \nabla \times (\nabla \times \underline{V}) = \hat{x} \left[\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) V_x \right] + \hat{y} \left[\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) V_y \right] \\ + \hat{z} \left[\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) V_z \right]$$

$$\nabla \cdot (\nabla \cdot \underline{V}) - \nabla \times (\nabla \times \underline{V}) = \hat{x} (\nabla^2 V_x) + \hat{y} (\nabla^2 V_y) + \hat{z} (\nabla^2 V_z) = \nabla^2 (\hat{x} V_x + \hat{y} V_y + \hat{z} V_z) = \nabla^2 \underline{V}$$

$$\nabla^2 \underline{V} = \nabla \cdot (\nabla \cdot \underline{V}) - \nabla \times (\nabla \times \underline{V}) = \hat{x} \nabla^2 V_x + \hat{y} \nabla^2 V_y + \hat{z} \nabla^2 V_z$$

4) If $\phi(x, y, z) = x^3 y^2 z$, find $\nabla \phi$ and $\nabla \cdot \nabla \phi$.

$$\text{Gradient: } \nabla \phi = \hat{x} \frac{\partial \phi}{\partial x} + \hat{y} \frac{\partial \phi}{\partial y} + \hat{z} \frac{\partial \phi}{\partial z}$$

$$\nabla \phi = \hat{x} \left(\frac{\partial}{\partial x} (x^3 y^2 z) \right) + \hat{y} \left(\frac{\partial}{\partial y} (x^3 y^2 z) \right) + \hat{z} \left(\frac{\partial}{\partial z} (x^3 y^2 z) \right)$$

$$\nabla \phi = \hat{x} (3x^2 y^2 z) + \hat{y} (2x^3 y z) + \hat{z} (x^3 y^2)$$

$$\nabla \phi = \hat{x} (3x^2 y^2 z) + \hat{y} (2x^3 y z) + \hat{z} (x^3 y^2)$$

$$\nabla = \hat{x} \frac{\partial}{\partial x} + \hat{y} \frac{\partial}{\partial y} + \hat{z} \frac{\partial}{\partial z}$$

$$\nabla \cdot \nabla \phi = \frac{\partial}{\partial x} (3x^2 y^2 z) + \frac{\partial}{\partial y} (2x^3 y z) + \frac{\partial}{\partial z} (x^3 y^2)$$

$$\nabla \cdot \nabla \phi = 6xy^2z + 2x^3z + 0$$

$$\nabla \cdot \nabla \phi = 6xy^2z + 2x^3z$$

- 5) Given that $\underline{E}(x, y, z, t) = 2x\hat{x} + y\hat{y} + 3z\hat{z}$ is a field that exists in space, does $\underline{B}(x, y, z, t)$ vary with time? Justify your answer. Repeat for $\underline{E}(x, y, z, t) = 2y\hat{x} + z\hat{y} + 3x\hat{z}$.

Using Faraday's law, we can determine whether $\underline{B}$ varies with time.

$$\nabla \times \underline{E} = -\frac{\partial \underline{B}}{\partial t}$$

If $\nabla \times \underline{E} = 0$, then $\frac{\partial \underline{B}}{\partial t} = 0$. This means $\underline{B}$ does not vary with time.

If $\nabla \times \underline{E} \neq 0$, then $\frac{\partial \underline{B}}{\partial t} \neq 0$. This means $\underline{B}$ varies with time.

$$\nabla = \hat{x} \frac{\partial}{\partial x} + \hat{y} \frac{\partial}{\partial y} + \hat{z} \frac{\partial}{\partial z}$$

$$\nabla \times \underline{E} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2x & y & 3z \end{vmatrix} = \hat{x} \left[\frac{\partial}{\partial y}(3z) - \frac{\partial}{\partial z}(y) \right] + \hat{y} \left[\frac{\partial}{\partial z}(2x) - \frac{\partial}{\partial x}(3z) \right] + \hat{z} \left[\frac{\partial}{\partial x}(y) - \frac{\partial}{\partial y}(2x) \right]$$

$$\nabla \times \underline{E} = \hat{x}[0-0] + \hat{y}[0-0] + \hat{z}[0-0]$$

$$\nabla \times \underline{E} = \underline{0}$$

No, $\underline{B}(x, y, z, t)$ does not vary with time.

6) Given that $\underline{D}(x, y, z, t) = 2xy^2\hat{x} + 3x\hat{y} - x^2y\hat{z}$ in some region, what is the charge density in that region?

To find the charge density of that region, we can use the Electric Gauss law.

$$\nabla \cdot \underline{D} = \rho_v$$

$$\nabla = \hat{x} \frac{\partial}{\partial x} + \hat{y} \frac{\partial}{\partial y} + \hat{z} \frac{\partial}{\partial z}$$

$$\nabla \cdot \underline{D} = \frac{\partial}{\partial x}(2xy^2) + \frac{\partial}{\partial y}(3x) + \frac{\partial}{\partial z}(-x^2y)$$

$$\nabla \cdot \underline{D} = 2y^2 + 0 + 0$$

$$\nabla \cdot \underline{D} = 2y^2$$

$$\rho_v = 2y^2$$

$$\rho_v = 2y^2$$

The charge density, ρ_v , in that region is $2y^2$.

7) For the following sinusoidal electric fields in free space, find the magnetic field $\underline{H}(x, y, z, t)$.

(a) $\underline{E}(x, y, z, t) = E_0 (\underline{x}\hat{x} + \cos(2\pi z)\hat{y} + 2z\hat{z}) \sin \omega t$

(b) $\underline{E}(x, y, z, t) = E_0 \hat{z} \sin[2\pi \times 10^6 t + 2\pi(2x + 3y)]$

(c) $\underline{E}(x, y, z, t) = \hat{y} E_0 x^2 \cos \omega t$.

Faraday's Law: $\nabla \times \underline{E} = -\frac{\partial \underline{B}}{\partial t}$

Constitutive Relations in Free Space

$\underline{B} = \mu_0 \underline{H}$, (μ_0 = permeability), $\mu_0 = 4\pi \times 10^{-7} \text{ [H/m]}$

$\nabla \times \underline{E} = -\frac{\partial \underline{B}}{\partial t} \quad \rightarrow \quad \frac{\partial \underline{H}}{\partial t} = -\frac{1}{\mu_0} (\nabla \times \underline{E})$

$\nabla \times \underline{E} = -\frac{\partial}{\partial t} (\mu_0 \underline{H}) \quad , \quad \underline{H} = \int -\frac{1}{\mu_0} (\nabla \times \underline{E}) dt$

$\nabla \times \underline{E} = -\mu_0 \frac{\partial \underline{H}}{\partial t}$

(a) $\underline{E}(x, y, z, t) = \hat{x} [E_0 x \sin(\omega t)] + \hat{y} [E_0 \cos(2\pi z) \sin(\omega t)] + \hat{z} [2E_0 z \sin(\omega t)]$

$\nabla = \hat{x} \frac{\partial}{\partial x} + \hat{y} \frac{\partial}{\partial y} + \hat{z} \frac{\partial}{\partial z}$

$\nabla \times \underline{E} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ E_0 x \sin(\omega t) & E_0 \cos(2\pi z) \sin(\omega t) & 2E_0 z \sin(\omega t) \end{vmatrix} = \hat{x} \left[\frac{\partial}{\partial y} (2E_0 z \sin(\omega t)) - \frac{\partial}{\partial z} (E_0 \cos(2\pi z) \sin(\omega t)) \right] \\ + \hat{y} \left[\frac{\partial}{\partial z} (E_0 x \sin(\omega t)) - \frac{\partial}{\partial x} (2E_0 z \sin(\omega t)) \right] \\ + \hat{z} \left[\frac{\partial}{\partial x} (E_0 \cos(2\pi z) \sin(\omega t)) - \frac{\partial}{\partial y} (E_0 x \sin(\omega t)) \right]$

$\nabla \times \underline{E} = \hat{x} [0 - (-2\pi E_0 \sin(2\pi z) \sin(\omega t))] + \hat{y} [0 - 0] + \hat{z} [0 - 0]$

$\nabla \times \underline{E} = \hat{x} (2\pi E_0 \sin(2\pi z) \sin(\omega t))$

$\underline{H} = \int -\frac{1}{\mu_0} [\hat{x} (2\pi E_0 \sin(2\pi z) \sin(\omega t))] dt = \hat{x} \left[-\frac{1}{\mu_0} (2\pi E_0 \sin(2\pi z)) \right] \cdot \int \sin(\omega t) dt$

$\underline{H} = \hat{x} \left[-\frac{2\pi E_0}{\mu_0} \sin(2\pi z) \right] \cdot \left[-\frac{\cos(\omega t)}{\omega} + H_0 \right]$

$\underline{H} = \hat{x} \left[\frac{2\pi E_0}{\mu_0 \omega} \sin(2\pi z) \cos(\omega t) \right] + H_0 \quad \leftarrow \text{We can assume there is no DC magnetic field component.}$

$\underline{H} = \hat{x} \left[\frac{2\pi E_0}{\mu_0 \omega} \sin(2\pi z) \cos(\omega t) \right]$

$$\underline{H}(x, y, z, t) = \hat{x} \left[\frac{2\pi E_0}{\mu_0 \omega} \sin(2\pi z) \cos(\omega t) \right]$$

Part (a)
ANSWER

$$(b) \underline{E}(x, y, z, t) = E_0 \hat{z} \sin[2\pi \times 10^6 t + 2\pi(2x + 3y)]$$

$$\nabla = \hat{x} \frac{\partial}{\partial x} + \hat{y} \frac{\partial}{\partial y} + \hat{z} \frac{\partial}{\partial z}$$

$$\nabla \times \underline{E} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & 0 & E_0 \sin[2\pi \times 10^6 t + 4\pi x + 6\pi y] \end{vmatrix} = \hat{x} \left[\frac{\partial}{\partial y} (E_0 \sin(2\pi \times 10^6 t + 4\pi x + 6\pi y)) - \frac{\partial}{\partial z} (0) \right] + \hat{y} \left[\frac{\partial}{\partial z} (0) - \frac{\partial}{\partial x} (E_0 \sin(2\pi \times 10^6 t + 4\pi x + 6\pi y)) \right] + \hat{z} \left[\frac{\partial}{\partial x} (0) - \frac{\partial}{\partial y} (0) \right]$$

$$\nabla \times \underline{E} = \hat{x} [E_0 \cos(2\pi \times 10^6 t + 4\pi x + 6\pi y) \cdot (6\pi) - 0] + \hat{y} [0 - E_0 \cos(2\pi \times 10^6 t + 4\pi x + 6\pi y) \cdot (4\pi)] + \hat{z} [0 - 0]$$

$$\nabla \times \underline{E} = \hat{x} (-6\pi E_0 \cos(2\pi \times 10^6 t + 4\pi x + 6\pi y)) + \hat{y} (-4\pi E_0 \cos(2\pi \times 10^6 t + 4\pi x + 6\pi y))$$

$$\underline{H} = \int -\frac{1}{\mu_0} (\nabla \times \underline{E}) dt = \int -\frac{1}{\mu_0} [\hat{x} (6\pi E_0 \cos(2\pi \times 10^6 t + 4\pi x + 6\pi y)) + \hat{y} (-4\pi E_0 \cos(2\pi \times 10^6 t + 4\pi x + 6\pi y))] dt$$

$$\underline{H} = -\frac{2\pi E_0}{\mu_0} \cdot \int [\hat{x} (3 \cos(2\pi \times 10^6 t + 4\pi x + 6\pi y)) + \hat{y} (-2 \cos(2\pi \times 10^6 t + 4\pi x + 6\pi y))] dt$$

$$\underline{H} = -\frac{2\pi E_0}{\mu_0} \cdot \left[\hat{x} \left(\frac{3 \sin(2\pi \times 10^6 t + 4\pi x + 6\pi y)}{2\pi \times 10^6} \right) + \hat{y} \left(-\frac{2 \sin(2\pi \times 10^6 t + 4\pi x + 6\pi y)}{2\pi \times 10^6} \right) + H_0 \right]$$

$$\underline{H} = \hat{x} \left[-\frac{3E_0}{\mu_0 \times 10^6} \sin(2\pi \times 10^6 t + 4\pi x + 6\pi y) \right] + \hat{y} \left[\frac{2E_0}{\mu_0 \times 10^6} \sin(2\pi \times 10^6 t + 4\pi x + 6\pi y) \right] + H_0 \quad \text{assume no DC component}$$

Part (b)
ANSWER $\rightarrow \underline{H}(x, y, z, t) = \hat{x} \left[-\frac{3E_0}{\mu_0 \times 10^6} \sin(2\pi \times 10^6 t + 4\pi x + 6\pi y) \right] + \hat{y} \left[\frac{2E_0}{\mu_0 \times 10^6} \sin(2\pi \times 10^6 t + 4\pi x + 6\pi y) \right]$

$$(c) \underline{E}(x, y, z, t) = \hat{z} E_0 x^2 \cos(\omega t), \quad \nabla = \hat{x} \frac{\partial}{\partial x} + \hat{y} \frac{\partial}{\partial y} + \hat{z} \frac{\partial}{\partial z}$$

$$\nabla \times \underline{E} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & E_0 x^2 \cos(\omega t) & 0 \end{vmatrix} = \hat{x} \left[\frac{\partial}{\partial y} (0) - \frac{\partial}{\partial z} (E_0 x^2 \cos(\omega t)) \right] + \hat{y} \left[\frac{\partial}{\partial z} (0) - \frac{\partial}{\partial x} (0) \right] + \hat{z} \left[\frac{\partial}{\partial x} (E_0 x^2 \cos(\omega t)) - \frac{\partial}{\partial y} (0) \right]$$

Assume no DC component

$$\nabla \times \underline{E} = \hat{x} [0 - 0] + \hat{y} [0 - 0] + \hat{z} [2E_0 x \cos(\omega t) - 0] = \hat{z} (2E_0 x \cos(\omega t))$$

$$\underline{H} = \int -\frac{1}{\mu_0} (\nabla \times \underline{E}) dt = \int -\frac{1}{\mu_0} [\hat{z} (2E_0 x \cos(\omega t))] dt = -\frac{2E_0 x}{\mu_0} \int \hat{z} \cos(\omega t) dt = -\frac{2E_0 x}{\mu_0} \cdot \left[\hat{z} \left(\frac{\sin(\omega t)}{\omega} \right) + H_0 \right]$$

$$\underline{H} = \hat{z} \left(-\frac{2E_0 x}{\mu_0 \omega} \sin(\omega t) \right)$$

$$\underline{H}(x, y, z, t) = \hat{z} \left(-\frac{2E_0 x}{\mu_0 \omega} \sin(\omega t) \right)$$

8) If the phasor-domain fields in a region of space are

$$\underline{E} = (\hat{x} - j\hat{y})e^{-j(2\pi)z}$$

$$\underline{H} = 0.002654(\hat{x} - j\hat{y})e^{-j(2\pi)z},$$

find the complex Poynting vector $\underline{S}$ and the time average Poynting vector $\langle \underline{S}(t) \rangle$.

Complex Poynting vector $\underline{S}$: $\underline{S} \equiv \frac{1}{2}(\underline{E} \times \underline{H}^*)$

$$\underline{E} = (\hat{x} - j\hat{y})e^{-j(2\pi)z} = \hat{x}(e^{-j(2\pi)z}) + \hat{y}(-je^{-j(2\pi)z})$$

$$\underline{H}^* = 0.002654(\hat{x} + j\hat{y})e^{j(2\pi)z} = \hat{x}(0.002654e^{j(2\pi)z}) + \hat{y}(j0.002654e^{j(2\pi)z})$$

$$\underline{E} \times \underline{H}^* = \begin{vmatrix} \hat{x} & \hat{y} \\ e^{-j(2\pi)z} & -je^{-j(2\pi)z} \\ 0.002654e^{j(2\pi)z} & j0.002654e^{j(2\pi)z} \end{vmatrix} = \hat{z} \left[e^{-j(2\pi)z} (j0.002654e^{j(2\pi)z}) - (-je^{-j(2\pi)z})(0.002654e^{j(2\pi)z}) \right]$$

$$\underline{E} \times \underline{H}^* = \hat{z} \left[j0.002654e^{j(2\pi)z-j(2\pi)z} + j0.002654e^{j(2\pi)z-j(2\pi)z} \right] = \hat{z} \left[j0.002654e^0 + j0.002654e^0 \right]$$

$$\underline{E} \times \underline{H}^* = \hat{z} (j0.005308)$$

$$\underline{S} = \frac{1}{2}(\underline{E} \times \underline{H}^*) = \frac{1}{2}[\hat{z}(j0.005308)] = \hat{z}(j0.002654)$$

$$\langle \underline{S}(t) \rangle = \langle \underline{E}(t) \times \underline{H}(t) \rangle = \frac{1}{2} \text{Re}(\underline{E} \times \underline{H}^*)$$

$\underline{E} \times \underline{H}^*$ has no real component

$$\langle \underline{S}(t) \rangle = \frac{1}{2}(0) = 0 \text{ [W/m}^2\text{]}$$

$$\underline{S} = \frac{1}{2}(\underline{E} \times \underline{H}^*) = \hat{z}(j0.002654)$$

$$\langle \underline{S}(t) \rangle = 0 \text{ [W/m}^2\text{]}$$

- 9) Consider a coaxial cable of inner radius a and outer radius b , filled with a lossless dielectric material. The phasor-domain fields inside of the cable ($a < \rho < b$) for a time-harmonic wave traveling down the cable are given in cylindrical coordinates as

$$\underline{E} = \underline{\hat{\rho}} \left(\frac{1}{\rho} \right) e^{-jkz} \quad \underline{\hat{\rho}} \times \underline{\hat{\phi}} = \underline{\hat{z}}$$

$$\underline{H} = \underline{\hat{\phi}} \left(\frac{1}{\rho} \right) \frac{1}{\eta_0 \sqrt{\epsilon_r}} e^{-jkz} \quad \underline{H}^* = \underline{\hat{\phi}} \left(\frac{1}{\rho} \right) \frac{1}{\eta_0 \sqrt{\epsilon_r}} e^{jkz}$$

Calculate the total time-average power (in Watts) flowing in the z direction down the cable. (Note: The real-valued wavenumber k is the wavenumber of the medium filling the coax, $k = \omega \sqrt{\mu \epsilon}$, but its value should not affect your answer.).

Time-average Poynting Vector: $\langle S(t) \rangle = \langle \underline{E}(t) \times \underline{H}(t) \rangle = \frac{1}{2} \text{Re}(\underline{E} \times \underline{H}^*)$

Time-average power flowing: $P_{\text{avg}} = \iint_S \langle S(t) \rangle \cdot d\mathbf{A}$, $dA = \rho d\rho d\phi$

$$\underline{E} \times \underline{H}^* = \begin{vmatrix} \underline{\hat{\rho}} & \underline{\hat{\phi}} & \underline{\hat{z}} \\ \left(\frac{1}{\rho}\right)e^{-jkz} & 0 & 0 \\ 0 & \left(\frac{1}{\rho}\right)\frac{1}{\eta_0 \sqrt{\epsilon_r}} e^{jkz} & 0 \end{vmatrix} = \underline{\hat{\rho}} \left[0(0) - 0\left(\frac{1}{\rho}\right)\frac{1}{\eta_0 \sqrt{\epsilon_r}} e^{jkz} \right] + \underline{\hat{\phi}} \left[0(0) - \left(\frac{1}{\rho}\right)e^{-jkz}(0) \right] + \underline{\hat{z}} \left[\left(\frac{1}{\rho}\right)e^{-jkz} \cdot \left(\frac{1}{\rho}\right)\frac{1}{\eta_0 \sqrt{\epsilon_r}} e^{jkz} - 0(0) \right]$$

$$\underline{E} \times \underline{H}^* = \underline{\hat{z}} \left(\frac{1}{\rho^2 \eta_0 \sqrt{\epsilon_r}} e^{jkz-jkz} \right) = \underline{\hat{z}} \left(\frac{1}{\rho^2 \eta_0 \sqrt{\epsilon_r}} \right)$$

$$\langle S(t) \rangle = \frac{1}{2} \text{Re}(\underline{E} \times \underline{H}^*) = \underline{\hat{z}} \left(\frac{1}{2 \rho^2 \eta_0 \sqrt{\epsilon_r}} \right)$$

$$P_{\text{avg}} = \iint_S \underline{\hat{z}} \left(\frac{1}{2 \rho^2 \eta_0 \sqrt{\epsilon_r}} \right) dA = \int_0^{2\pi} \int_a^b \underline{\hat{z}} \left(\frac{1}{2 \rho^2 \eta_0 \sqrt{\epsilon_r}} \right) \cdot \rho d\rho d\phi = \underline{\hat{z}} \left(\frac{1}{2 \eta_0 \sqrt{\epsilon_r}} \right) \cdot \int_0^{2\pi} \int_a^b \frac{1}{\rho} d\rho d\phi$$

$$P_{\text{avg}} = \underline{\hat{z}} \left(\frac{1}{2 \eta_0 \sqrt{\epsilon_r}} \right) \cdot \int_0^{2\pi} (\ln(\rho)) \Big|_a^b d\phi = \underline{\hat{z}} \left(\frac{1}{2 \eta_0 \sqrt{\epsilon_r}} \right) \cdot \int_0^{2\pi} (\ln(b) - \ln(a)) d\phi$$

$$P_{\text{avg}} = \underline{\hat{z}} \left(\frac{1}{2 \eta_0 \sqrt{\epsilon_r}} \right) \cdot \int_0^{2\pi} \ln\left(\frac{b}{a}\right) d\phi = \underline{\hat{z}} \left(\frac{\ln\left(\frac{b}{a}\right)}{2 \eta_0 \sqrt{\epsilon_r}} \right) \cdot \int_0^{2\pi} d\phi = \underline{\hat{z}} \left(\frac{\ln\left(\frac{b}{a}\right)}{2 \eta_0 \sqrt{\epsilon_r}} \right) \cdot (\phi) \Big|_0^{2\pi}$$

$$P_{\text{avg}} = \underline{\hat{z}} \left(\frac{\ln\left(\frac{b}{a}\right)}{2 \eta_0 \sqrt{\epsilon_r}} \right) \cdot (2\pi - 0) = \underline{\hat{z}} \left(\frac{\pi \ln\left(\frac{b}{a}\right)}{2 \eta_0 \sqrt{\epsilon_r}} \right)$$

$$P_{\text{avg}} = \underline{\hat{z}} \left(\frac{\pi \ln\left(\frac{b}{a}\right)}{2 \eta_0 \sqrt{\epsilon_r}} \right)$$